
Stop Polling: Event-Driven AI Agents

1 Why polling loops break AI agents in production

2 The architecture shift to real-time event triggers

The Polling Bottleneck

- 1 Constant 'any updates?' checks waste compute cycles and hit API rate limits.
- 2 Adds up to 90% latency to critical real-time workflows.

The Event-Driven Flow

- 1 Producers emit events to a lightweight message broker.
- 2 Filter layer routes payloads only to subscribed agent workers.
- 3 Agents wake up on-demand with pre-loaded context.

The Worker Pattern

- 1 Subscribe directly to specific business topics instead of scheduling timers.

main.py

```
def agent_worker(event_type, handler):
    pubsub = redis_client.pubsub()
    pubsub.subscribe(event_type)
    for msg in pubsub.listen():
        if msg["type"] == "message":
            handler(msg["data"])
```

Real Production Impact

- 1 Zero idle compute costs while waiting for work.
- 2 Instant sub-second response times for agent actions.

Summary & Next Steps

- 1 Use Redis or SQS before over-engineering with Kafka.
- 2 Make handlers idempotent to safely handle duplicate events.

